

Geometry-Driven Poisson Flow Model of Cortical Manifolds

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Data Availability Statement

The imaging data analyzed in this study were obtained from the Juvenile Myoclonic Epilepsy Connectome Project (JMECP) collected at University of Wisconsin-Madison. Due to patient privacy and institutional restrictions, raw imaging data are not publicly available. De-identified simulated data and complete analysis code supporting the findings of this study will be publicly available on GitHub upon publication.

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Conflict of Interest

The authors declare that they have no competing financial interests or personal relationships that could have influenced the work reported in this paper.

Ethics Approval Statement

All procedures involving human participants were conducted in accordance with the ethical standards of the institutional research committee. The Juvenile Myoclonic Epilepsy Connectome Project was approved by the Institutional Review Board of the University of Wisconsin–Madison and Washington University in St. Louis.

Patient Consent Statement

Written informed consent was obtained from all participants. For participants under 18 years of age, written parental consent and participant assent were obtained prior to enrollment.

GEOMETRY-DRIVEN POISSON FLOW MODEL OF CORTICAL MANIFOLDS

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ABSTRACT

Cortical folding reflects coordinated neurodevelopmental processes and is increasingly recognized as a sensitive marker of neurological disease. However, most existing analyses rely on indirect scalar summaries that do not explicitly model folding geometry itself. In juvenile myoclonic epilepsy (JME), a common genetic epilepsy, cortical abnormalities are often subtle, spatially distributed, and difficult to detect using conventional morphometric measures. We introduce a Poisson-equation-based framework that models cortical folding as a geometry-driven flow derived from mean curvature on the cortical manifold. By treating folding patterns as a stationary source-sink structure, the proposed approach yields a smooth, globally balanced potential field whose surface gradient defines a physically interpretable flux. This framework enables spatially coherent analysis of sulcal-gyral folding organization and provides a principled representation of geometry-driven cortical structure in JME.

1. INTRODUCTION

Cortical folding arises from a complex interplay between neurodevelopmental processes, mechanical constraints, and differential growth, and its organization is closely linked to both normal brain function and neurological disease [1, 2]. Accordingly, abnormalities in sulcal and gyral morphology have been reported across a wide range of conditions, including epilepsy [3], neurodevelopmental disorders [4], and neurodegenerative diseases [5], indicating that folding patterns encode clinically meaningful information beyond gross cortical shape.

Juvenile myoclonic epilepsy (JME) is a genetic epilepsy with onset in adolescence and a strong neurodevelopmental component. Although conventional structural MRI is typically unremarkable, converging evidence suggests that JME is associated with subtle and spatially distributed alterations in brain organization. Prior morphometric studies have reported distributed cortical abnormalities in JME, particularly involving fronto-central and higher-order association regions [3, 40], consistent with a network-level neurodevelopmental

disorder rather than focal cortical pathology, while our recent volumetric study [6] demonstrates selective alterations in motor-associated thalamic nuclei and corticothalamic circuits, supporting a distributed neurodevelopmental network disorder rather than focal cortical pathology. Together, these findings indicate that JME reflects altered neurodevelopment of distributed brain networks, rather than localized cortical abnormalities.

Despite this growing recognition of JME as a *distributed* neurodevelopmental network disorder [3, 6], most existing structural imaging studies continue to rely on indirect summary measures such as cortical thickness, gyrification index, or regional volumes. These measures primarily quantify local gray-matter morphometry and provide limited information about the intrinsic geometry and spatial organization of sulcal-gyral folding patterns themselves [7, 8, 9]. While informative for detecting focal or regional differences, such scalar descriptors reduce complex folding architecture to isolated measurements and therefore cannot capture how sulci and gyri are arranged as a coordinated, continuous structure across the cortical surface. This limitation is particularly consequential in JME, where abnormalities are hypothesized to reflect large-scale developmental alterations spanning fronto-central and association networks rather than localized changes in individual cortical regions. These considerations motivate the need for a new frameworks that explicitly model cortical folding as a geometric object, enabling spatially coherent and biologically interpretable characterization of folding organization in JME.

These observations motivate the need for quantitative frameworks that can characterize folding patterns in a principled, spatially coherent manner and link local geometric organization to disease-related processes. However, most existing studies of cortical folding rely on indirect summary measures—such as cortical thickness and gyrification index—which primarily characterize local gray-matter morphometry rather than the intrinsic shape and spatial organization of sulcal-gyral folding patterns themselves [7, 8, 9]. Although these measures are informative, they reduce complex folding geometry to isolated scalar descriptors and therefore cannot capture how sulci and gyri are organized as a coordinated, continuous structure across the cortical surface. This limitation is particularly consequential in TLE, where

folding abnormalities are often spatially distributed across the temporal lobe and peri-temporal regions and may reflect large-scale geometric reorganization rather than focal changes in individual sulci [10]. A framework that explicitly models folding patterns as shapes on the cortical manifold is therefore needed.

In this study, we model cortical folding as a geometry-driven flow governed by a Poisson equation acting on the mean curvature field. Conventional surface-based approaches typically represent scalar data on cortical manifolds using heat diffusion or heat kernel formulations [11, 12], in which signals evolve under a parabolic partial differential equation. While effective for stabilizing noise, these models describe cortical geometry through time-dependent diffusion, leading to attenuation of spatial contrast and blurring of coordinated folding patterns as diffusion time increases [12, 13, 14]. Instead of diffusing curvature data forward in time, the proposed new approach formulates folding geometry directly as a source-sink pattern and recovers a smooth potential field by solving a Poisson equation on the cortical surface. This yields a globally balanced, spatially coherent representation that suppresses high-frequency noise while preserving relative geometric organization, and naturally induces a surface flux field given by the gradient of the potential, providing a physically interpretable, directional descriptor of geometry-driven cortical folding.

2. METHODS

2.1. Imaging Data and Preprocessing

Participants were drawn from the Juvenile Myoclonic Epilepsy Connectome Project (JMECP), a prospective study conducted at the University of Wisconsin Hospital and Clinics [6]. Detailed screening procedures and inclusion and exclusion criteria are reported in [6]. The present study included 61 individuals with JME (mean age 19.72 ± 3.70 years) and 39 healthy controls (mean age 21.44 ± 2.35 years) with no history of neurological or psychiatric disorders. There were 22 males and 39 females in JME, while the control group included 19 males and 20 females. There was no significant difference in sex distribution between the groups (Fisher’s exact test [15], $p = 0.22$). However, JME and control groups differed significantly in age (two-sample t-test [16], $p = 0.011$). Thus, age and sex were included as covariates in all group-level analyses.

All participants underwent structural MRI scanning on GE MR750 3T clinical scanners using a 32-channel phased-array head coil. High-resolution T1-weighted images were acquired as part of a comprehensive epilepsy imaging protocol. Intensity inhomogeneity was corrected using the N4 bias correction algorithm prior to further processing [17]. Cortical reconstruction was performed using the standard reconstruction pipeline in FreeSurfer [18], including motion correction,

intensity normalization, Talairach transformation, skull stripping, and automated segmentation.

Individual cortical meshes were extracted following the FreeSurfer processing stream, yielding triangulated surfaces with a fixed topology of 32,492 vertices and 64,980 faces per hemisphere. These subject-specific cortical surfaces were subsequently mapped into MNI space using FreeSurfer’s surface-based registration framework [19], establishing vertex-wise correspondence across subjects while preserving the intrinsic cortical geometry.

2.2. Estimating cortical folding pattern

Let $\mathcal{M}$ denote the closed cortical manifold discretized as a triangulated surface mesh. Cortical folding was represented by the vertexwise mean curvature field $h(x)$ defined at each vertex x of the mesh. Mean curvature was estimated using a local quadratic surface fitting approach based on first-ring neighboring vertices [11, 20]. In a local coordinate $x = (x_1, x_2)$, the surface was estimated as the quadratic form

$$f(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \frac{1}{2} \beta_3 x_1^2 + \beta_4 x_1 x_2 + \frac{1}{2} \beta_5 x_2^2$$

using the least squares fit. The first and second fundamental forms were computed as

$$g = \begin{pmatrix} 1 + \beta_1^2 & \beta_1 \beta_2 \\ \beta_1 \beta_2 & 1 + \beta_2^2 \end{pmatrix}, \quad h = \begin{pmatrix} \beta_3 & \beta_4 \\ \beta_4 & \beta_5 \end{pmatrix}.$$

Subsequently, the mean curvature at vertex x was then estimated as

$$h(x) = \frac{1}{2} \text{tr}(g^{-1}h).$$

By convention, positive curvatures correspond to sulcal fundi, which we interpret as sources, while negative curvatures correspond to gyral crowns, interpreted as sinks. This scalar field therefore encodes the local folding polarity of the cortex in a simple and intuitive way.

For subsequent analysis, we normalize $h(v)$ by subtracting its area-weighted average,

$$h(x) \leftarrow h(x) - \frac{1}{\text{Area}(\mathcal{M})} \int_{\mathcal{M}} h(x) d\mu(x),$$

which enforces the zero-mean condition

$$\int_{\mathcal{M}} h(x) d\mu(x) = 0, \quad (1)$$

which is required for a well-posed Poisson formulation on a closed manifold.

Intuitively, removing the global offset ensures that subsequent computations capture only relative, spatially varying folding patterns rather than being influenced by an arbitrary baseline curvature. As a result, the derived potential and flow fields can be interpreted as geometry-driven interactions

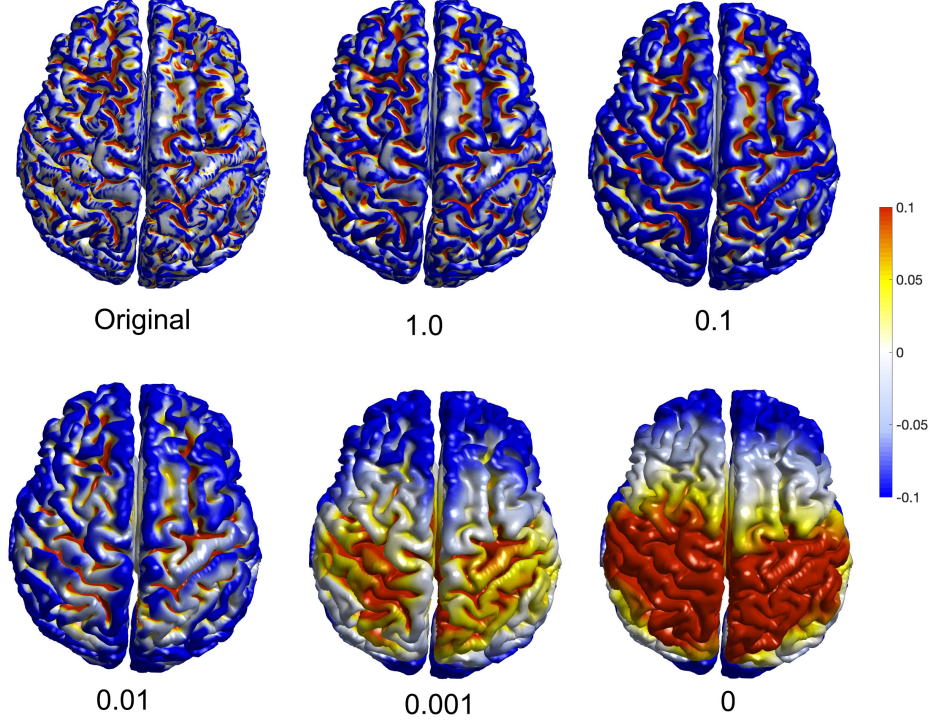


Fig. 1. Solutions of the regularized Poisson equation. When $\lambda = 0$, the formulation reduces to the standard Poisson equation, yielding solutions dominated by global modes and excessive smoothing. As λ increases, regularization suppresses long-range effects and enhances spatial locality. At very small regularization (e.g., $\lambda = 0.001$), the solution is primarily driven by intrinsic folding geometry, producing prominent structure around deeply and consistently folded regions such as primary sensorimotor cortex. In this study, we used $\lambda = 0.1$, which provides a balance between noise reduction and preservation of folding pattern.

between sulcal sinks and gyral sources on the cortical surface. Sulcal regions exhibit positive curvature values (around $+0.1$), whereas gyral crowns exhibit negative values (around -0.1), such that the sign of h indicates whether a local region is predominantly sulcal or gyral (Fig. 1, top left).

Sulcal cortex is consistently thinner than gyral cortex across much of the human brain [21, 22], a pattern that has been linked to mechanical tension, differential growth, and folding mechanics during development [1]. Classical models of cortical development suggest that neuronal number per unit surface area is approximately conserved across cortex [23, 24]. Under this assumption, reduced cortical thickness in sulcal regions implies higher neuronal packing density per unit volume relative to gyral regions, reflecting fundamental differences in cortical geometry rather than localized pathology. Consequently, the signed curvature field h and its derived potential provide a geometrically grounded representation that is closely tied to known microstructural and developmental differences between sulcal and gyral cortex.

2.3. Poisson model of cortical folding

Curvature estimates are known to be sensitive to discretization error, local mesh irregularity, and high-frequency geo-

metric noise [25]. To reduce the influence of such noise on downstream analyses, we smooth the folding pattern.

Let $\mathcal{L} = \nabla \cdot \nabla$ denote the Laplace–Beltrami operator on the closed cortical manifold $\mathcal{M}$, defined as the *divergence of the gradient* [26]. Given the area-normalized signed mean curvature field h (positive in sulci and negative in gyri), a natural smoothing strategy is to recover a potential field u whose Laplacian matches h ,

$$\mathcal{L}u = h \quad \text{on } \mathcal{M}. \quad (2)$$

Unlike the Euclidean setting, the Poisson equation (2) on a closed manifold does not automatically admit a solution. A solution exists only when the source field h satisfies the zero-mean condition (1):

Theorem 1 (Solvability on a closed manifold). *For a closed Riemannian manifold $\mathcal{M}$ without boundary, the Poisson equation*

$$\mathcal{L}u = h \quad \text{on } \mathcal{M}$$

admits a solution u if and only if h satisfies the zero-mean condition

$$\int_{\mathcal{M}} h \, d\mu = 0.$$

When this condition holds, the solution is unique up to an additive constant.

Proof. Assume first that u solves $\mathcal{L}u = h$. Then from the divergence theorem on a closed manifold [27], we have

$$\int_{\mathcal{M}} h d\mu = \int_{\mathcal{M}} \mathcal{L}u d\mu = \int_{\mathcal{M}} \nabla \cdot (\nabla u) d\mu = 0.$$

Hence the zero-mean condition is necessary.

Conversely, assume that $\int_{\mathcal{M}} h d\mu = 0$. Let $\{(\phi_k, \gamma_k)\}_{k=0}^{\infty}$ be an orthonormal eigenbasis of $\mathcal{L}$, so that

$$\mathcal{L}\phi_k = \gamma_k \phi_k, \quad 0 = \gamma_0 < \gamma_1 \leq \gamma_2 \leq \dots,$$

and ϕ_0 is constant. Since ϕ_0 is constant, we have $\langle h, \phi_0 \rangle = 0$. Define

$$u = \sum_{k=1}^{\infty} \frac{\langle h, \phi_k \rangle}{\gamma_k} \phi_k,$$

which is well defined. Then we have

$$\mathcal{L}u = \sum_{k=1}^{\infty} \frac{\langle h, \phi_k \rangle}{\gamma_k} \mathcal{L}\phi_k = \sum_{k=1}^{\infty} \langle h, \phi_k \rangle \phi_k.$$

On the other hand, expanding h in the same basis yields

$$h = \sum_{k=0}^{\infty} \langle h, \phi_k \rangle \phi_k = \sum_{k=1}^{\infty} \langle h, \phi_k \rangle \phi_k,$$

Therefore, the given series u is a solution.

Finally, if u_1 and u_2 are two different solutions, then $\mathcal{L}(u_1 - u_2) = 0$ implies $u_1 - u_2$ is constant. Hence the solution is unique up to an additive constant. $\square$

Theorem 1 explains why the Poisson equation $\mathcal{L}u = h$ on a closed manifold can admit a solution only when the mean curvature field h has the zero mean.

The Poisson equation acts as a low-pass filter: inversion of the Laplace–Beltrami operator $\mathcal{L}$ strongly attenuates high-frequency components of h while emphasizing its large-scale source–sink structure. The resulting solution u can be interpreted as the smoothest scalar field whose second-order variations reproduce the observed folding polarity. However, on a closed manifold, this strong suppression of high-frequency modes can lead to excessive smoothing, whereby local geometric features are attenuated and global modes dominate the solution (Fig. 1, bottom right). To mitigate this effect, we adopt a regularized Poisson formulation [28],

$$\mathcal{L}u + \lambda u = h, \quad (3)$$

where $\lambda > 0$ is a small regularization parameter. This modification preserves smoothness while increasing spatial locality, allowing mesoscale folding structure to be retained. At very low regularization (e.g., $\lambda = 0.001$), the solution approaches

the standard Poisson limit, in which the potential is driven primarily by strong, spatially coherent curvature sources. In this regime, prominent structure naturally emerges around deeply and consistently folded regions, such as primary sensorimotor cortex, reflecting the intrinsic geometric organization of cortical folding. In the present study, we set $\lambda = 0.01$ to balance noise suppression with preservation of local folding detail (Fig. 1, bottom left).

2.4. Regularized Poisson model

The regularized Poisson equation admits a closed-form analytic solution in terms of the spectral decomposition of the Laplace–Beltrami operator.

Theorem 2 (Spectral solution of the regularized Poisson equation). *The regularized Poisson equation*

$$(\mathcal{L} + \lambda)u = h, \quad \lambda > 0,$$

admits a unique solution given by

$$u = \sum_{k=0}^{\infty} \frac{\langle h, \phi_k \rangle}{\gamma_k + \lambda} \phi_k, \quad (4)$$

where $\{(\phi_k, \gamma_k)\}_{k=0}^{\infty}$ are the eigenpairs satisfying

$$\mathcal{L}\phi_k = \gamma_k \phi_k, \quad 0 = \gamma_0 < \gamma_1 \leq \gamma_2 \leq \dots$$

Proof. Since $\{\phi_k\}$ is an orthonormal basis, both h and u admit the expansions

$$h = \sum_{k=0}^{\infty} \langle h, \phi_k \rangle \phi_k, \quad u = \sum_{k=0}^{\infty} \langle u, \phi_k \rangle \phi_k.$$

Then we have

$$(\mathcal{L} + \lambda)u = \sum_{k=0}^{\infty} (\gamma_k + \lambda) \langle u, \phi_k \rangle \phi_k.$$

Equating this expression with the expansion of h and exploiting orthonormality of $\{\phi_k\}$ gives, for each k ,

$$(\gamma_k + \lambda) \langle u, \phi_k \rangle = \langle h, \phi_k \rangle.$$

Solving for $\langle u, \phi_k \rangle$ and substituting back into the expansion of u yields the desired expansion. $\square$

Theorem 2 allows construction of the ground-truth synthetic data in later Validation. The spectral representation makes the role of the regularization parameter λ explicit. Modes with large eigenvalues γ_k , which correspond to high-frequency, spatially localized fluctuations, are strongly damped, whereas low-frequency modes that encode large-scale folding patterns are preserved but bounded. Consequently, the inverse operator $(\mathcal{L} + \lambda)^{-1}$ acts as a geometry-adapted low-pass filter on the cortical manifold, reducing

noise-driven curvature variations while preventing domination by global modes on a closed surface. Increasing λ progressively suppresses long-range source–sink interactions and yields more spatially localized smoothing, while $\lambda = 0$ recovers the classical Poisson limit dominated by global structure. This links the regularized Poisson model directly to diffusion-based operators such as heat-kernel smoothing, with λ controlling the effective diffusion scale [4].

Theorem 3 (Heat-kernel representation of the regularized Poisson solution). *The regularized Poisson equation (3) admits the Laplace transform as solution*

$$u(x) = \int_0^\infty e^{-\lambda t} K_t * h(x) dt,$$

where

$$K_t * h(x) = \int_{\mathcal{M}} K_t(x, y) h(y) d\mu(y),$$

is the convolution with heat kernel K_t .

Proof. Expand $h = \sum_{k=0}^\infty \langle h, \phi_k \rangle \phi_k$. Define the heat kernel convolution as a multiplication of matrix exponential [26]

$$K_t * h = e^{-t\mathcal{L}} h = \sum_{k=0}^\infty e^{-t\gamma_k} \langle h, \phi_k \rangle \phi_k.$$

The Laplace transform on heat kernel smoothing is then

$$v = \int_0^\infty e^{-\lambda t} K_t * h dt = \sum_{k=0}^\infty \langle h, \phi_k \rangle \phi_k \int_0^\infty e^{-(\lambda+\gamma_k)t} dt.$$

The integral is given by

$$\int_0^\infty e^{-(\lambda+\gamma_k)t} dt = \frac{1}{\lambda + \gamma_k}.$$

Therefore,

$$v = \sum_{k=0}^\infty \frac{\langle h, \phi_k \rangle}{\lambda + \gamma_k} \phi_k,$$

which is exactly the spectral solution (4) given in Theorem 2. $\square$

Theorem 3 makes explicit that the regularized Poisson solution can be interpreted as a Laplace transform, with respect to diffusion time t , of heat-kernel smoothing. Short diffusion times correspond to highly local smoothing, whereas long diffusion times encode increasingly global interactions on the manifold. The exponential weighting $e^{-\lambda t}$ suppresses these long-time contributions, so increasing λ progressively down-weights global diffusion and yields more spatially localized potentials. This effect corresponds to attenuation of high-frequency eigenmodes through the factor $(\gamma_k + \lambda)^{-1}$, while low-frequency modes are preserved but uniformly bounded.

Equivalently, the solution admits a compact operator-level representation via the *resolvent identity* [29],

$$(\mathcal{L} + \lambda)^{-1} = \int_0^\infty e^{-\lambda t} e^{-t\mathcal{L}} dt,$$

which yields

$$u = (\mathcal{L} + \lambda)^{-1} h = \int_0^\infty e^{-\lambda t} K_t * h dt.$$

This representation follows from the linearity of the involved operators and the spectral calculus for self-adjoint operators. It shows that the regularized Poisson solution can be viewed either as a geometry-adapted low-pass spectral filter or, equivalently, as a superposition of heat-kernel smoothings across diffusion scales [12]. In this sense, the regularization parameter λ directly controls the effective diffusion scale, linking Poisson-based smoothing to diffusion- and heat-kernel-based approaches on the cortical manifold.

2.5. Variational interpretation of energy-minimizing flow

The Poisson formulation admits a natural interpretation as an energy-minimization principle on the manifold $\mathcal{M}$. Consider the energy functional

$$\mathcal{E}_\lambda(u) = \int_{\mathcal{M}} \|\nabla u\|^2 d\mu + \lambda \int_{\mathcal{M}} u^2 d\mu - 2 \int_{\mathcal{M}} u h d\mu.$$

The first term $\int_{\mathcal{M}} \|\nabla u\|^2 d\mu$ is the Dirichlet energy, which penalizes spatial roughness of the potential and enforces smoothness with respect to the intrinsic geometry of the cortical manifold [11, 26]. On a closed surface, this term admits the equivalent quadratic form

$$\int_{\mathcal{M}} \|\nabla u\|^2 d\mu = \int_{\mathcal{M}} u \mathcal{L} u d\mu = \langle u, \mathcal{L} u \rangle,$$

where $\langle \cdot, \cdot \rangle$ is the inner product with respect to the area measure $d\mu$. Minimizing the Dirichlet energy suppresses high-frequency components of u associated with large eigenvalues of $\mathcal{L}$, while preserving low-frequency modes that encode large-scale folding organization. Consequently, smoothness is enforced intrinsically along the cortical manifold rather than in the ambient Euclidean space.

The second term $\int_{\mathcal{M}} u^2 d\mu$ is a zeroth-order Tikhonov regularization term with weight $\lambda > 0$ that controls the global amplitude of the solution and renders the inverse problem well posed on a closed manifold by removing the null space of the Laplace–Beltrami operator [28]. This term limits the dominance of low-frequency global modes present in the standard Poisson formulation and increases the spatial locality of the recovered potential, yielding a screened Poisson equation with a unique, stable solution.

The final term $2 \int_{\mathcal{M}} u h d\mu$ is a data-fidelity term that linearly couples the potential to the curvature-derived source-sink field h , enforcing consistency between the inferred potential and the observed sulcal-gyral polarity of cortical folding. Through the resulting Euler-Lagrange equation, regions of positive curvature act as effective sources and regions of negative curvature as sinks for the induced flow, ensuring that the energy-minimizing potential reflects the intrinsic folding geometry of the cortical surface.

Taking the first variation of $\mathcal{E}(u)$ with respect to u yields the Euler-Lagrange equation [28, 26], which is exactly the regularized Poisson equation (3).

Theorem 4 (Variational derivation). *The energy functional $\mathcal{E}_\lambda(u)$ admits a unique minimizer, which satisfies the Euler-Lagrange equation*

$$\mathcal{L}u + \lambda u = h.$$

Proof. We first examine how the energy changes under an infinitesimal perturbation of u . Let η be an arbitrary smooth test function on manifold $\mathcal{M}$ and define a perturbed function

$$u_\varepsilon = u + \varepsilon \eta$$

for a small scalar ε . A necessary condition for u to minimize the energy functional $\mathcal{E}_\lambda$ is that its first variation vanishes [30], i.e.,

$$\left. \frac{d}{d\varepsilon} \mathcal{E}_\lambda(u + \varepsilon \eta) \right|_{\varepsilon=0} = 0.$$

The derivative of the Dirichlet energy term with respect to ε is

$$\left. \frac{d}{d\varepsilon} \int_{\mathcal{M}} \|\nabla(u + \varepsilon \eta)\|^2 d\mu \right|_{\varepsilon=0} = 2 \int_{\mathcal{M}} \nabla u \cdot \nabla \eta d\mu.$$

From Green's formula on a closed manifold [27], we have

$$\int_{\mathcal{M}} \nabla u \cdot \nabla \eta d\mu = \int_{\mathcal{M}} \mathcal{L}u \eta d\mu.$$

The derivative of the second term gives

$$\left. \frac{d}{d\varepsilon} \lambda \int_{\mathcal{M}} (u + \varepsilon \eta)^2 d\mu \right|_{\varepsilon=0} = 2\lambda \int_{\mathcal{M}} u \eta d\mu.$$

The derivative of the data-fidelity term gives

$$\left. \frac{d}{d\varepsilon} 2 \int_{\mathcal{M}} (u + \varepsilon \eta) h d\mu \right|_{\varepsilon=0} = 2 \int_{\mathcal{M}} \eta h d\mu.$$

Combining the three derivatives, we obtain

$$\int_{\mathcal{M}} (\mathcal{L}u + \lambda u - h) \eta d\mu = 0.$$

Since the test function η is arbitrary, the integrand must vanish pointwise, yielding the Euler-Lagrange equation. $\square$

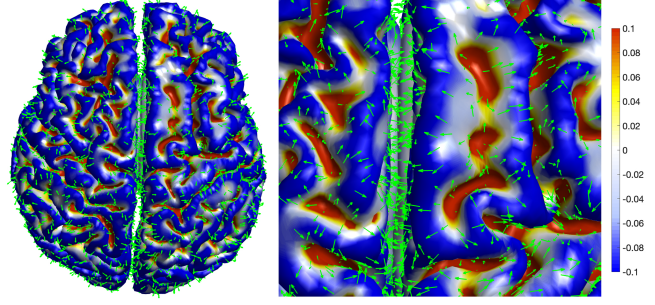


Fig. 2. Flux field defined as the negative surface gradient of the potential, pointing in the direction of steepest decrease of potential energy from sulcal regions (red) toward gyral regions (blue). Compared to the original curvature data, the resulting flux patterns are smoother and more coherent. Left: flux field visualized at 10% of vertices. Right: flux field visualized at 20% of vertices.

Within the variational framework, the Dirichlet energy defines an intrinsic energy landscape on the cortical manifold, shaped directly by the geometry of the surface. Cortical curvature modulates the potential field in a manner conceptually analogous to how spacetime curvature governs the motion of particles in general relativity [31]. Rather than invoking an external force acting through the ambient space, the present formulation captures geometry-driven structure intrinsic to the cortical surface itself, with the direction and magnitude of the induced flow emerging directly from local variations in curvature of the cortical sheet.

Since $\lambda > 0$, the functional $\mathcal{E}(u)$ is strictly convex, and therefore admits a unique minimizer. This variational formulation naturally induces a geometry-driven flow on the cortical surface. The surface gradient of the potential defines a tangential vector field

$$J = -\nabla u,$$

which points in the direction of steepest decrease of u and therefore represents the direction along which the energy $\mathcal{E}(u)$ decreases most rapidly. The regions of positive curvature act as sources and regions of negative curvature act as sinks for the induced flow. Consequently, the flux field J naturally propagates from sulcal fundi toward gyral crowns, yielding intrinsic, geometry-driven trajectories on the cortical manifold (Fig. 2). These flow field can be interpreted as energy-minimizing paths determined solely by the intrinsic geometry of cortical folding.

2.6. Numerical implementation

We solve the regularized Poisson equation on the triangulated cortical surface using the *cotangent* discretization of the Laplace-Beltrami operator [25, 7]. Let C and A denote the stiffness and mass matrices computed by the finite-element

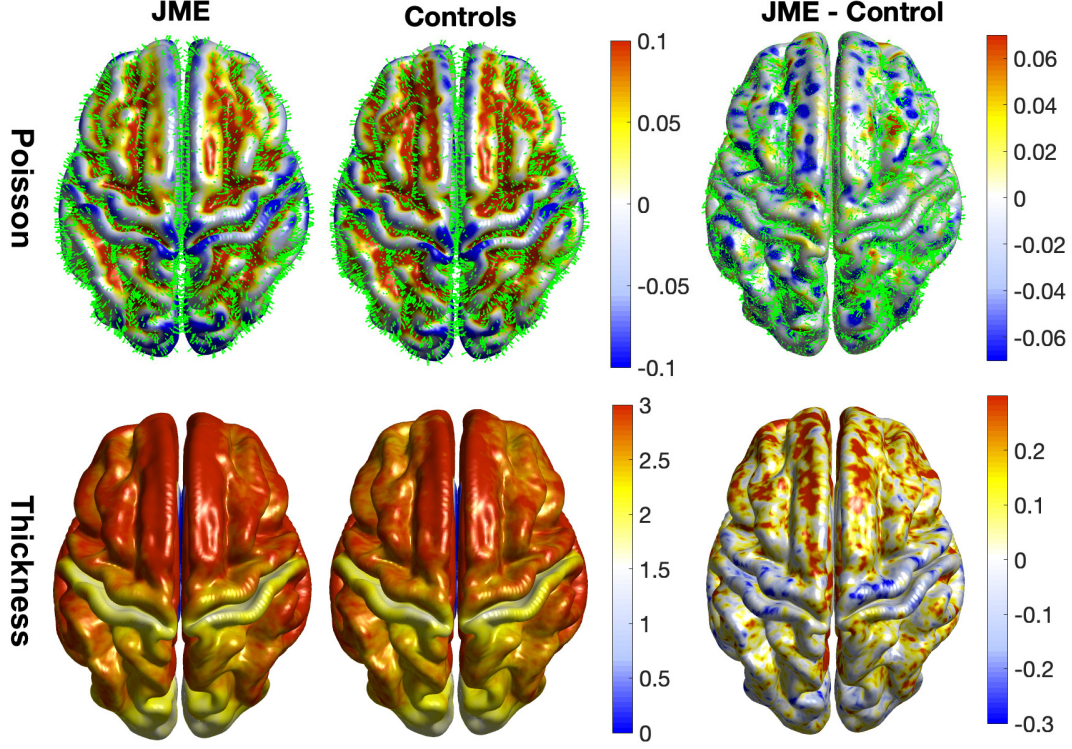


Fig. 3. Top: Group mean Poisson-derived folding potential and corresponding flux field (arrows) shown for JME, healthy controls, and their difference (JME–Control). Red indicates higher potential and blue indicates lower potential, with flux vectors depicting the mean geometry-driven flow on the cortical surface. **Bottom:** Corresponding group-mean cortical thickness maps (in millimeters) are shown to provide a laminar morphometric reference. In contrast to the Poisson-derived potential and flux, cortical thickness effects are visually subtler and less spatially discernible at the group level.

method [32]. The mass matrix A provides a finite-element approximation of local surface-area element:

$$\int_{\mathcal{M}} f(x) d\mu(x) \longrightarrow \mathbf{1}^\top A f,$$

where $\mathbf{1}$ denotes the constant vector of ones. Thus, a finite-element approximation of the surface area is

$$\text{Area}(\mathcal{M}) = \int_{\mathcal{M}} d\mu(x) \longrightarrow \mathbf{1}^\top A \mathbf{1}.$$

Further, A and C discretize surface inner products as

$$\langle u, v \rangle = \int_{\mathcal{M}} u v d\mu \longrightarrow u^\top A v, \quad (5)$$

$$\langle \nabla u, \nabla v \rangle = \int_{\mathcal{M}} \nabla u \cdot \nabla v d\mu \longrightarrow u^\top C v. \quad (6)$$

Under this discretization, the zero-mean condition (1) required for solvability of the Poisson equation, is enforced in discrete form as

$$\mathbf{1}^\top A h = 0.$$

This is achieved by subtracting the mean

$$\bar{h} = \frac{\int_{\mathcal{M}} h(x) d\mu(x)}{\int_{\mathcal{A}} 1 d\mu(x)} \longrightarrow \frac{\mathbf{1}^\top A h}{\mathbf{1}^\top A \mathbf{1}}.$$

We then discretized the regularized Poisson equation as

$$(C + \lambda A) u = A h.$$

We solve this linear system using a sparse Cholesky factorization [33]. The regularization parameter λ controls the trade-off between spatial locality and smoothness. Larger λ suppresses global low-frequency modes, yielding more localized folding patterns, while $\lambda \rightarrow 0$ recovers the classical Poisson solution dominated by large-scale structure. We set $\lambda = 0.1$ to balance noise suppression with preservation of local sulcal-gyral organization.

After solving the linear system, we apply two post-processing steps that do not alter the underlying geometry-driven flow or subsequent statistical analysis. First, we normalize the overall amplitude of u using the surface-averaged L^2 -norm so that its global amplitude matches that of the curvature field h as follows. The surface-averaged L^2 -norm $\|f\|^2$, which is equivalent to the root-mean-square error (RMSE) over the surface, is discretized as

$$\|f\|^2 = \frac{1}{\text{Area}(\mathcal{M})} \int_{\mathcal{M}} f(x)^2 d\mu(x) \longrightarrow \frac{f^\top A f}{\mathbf{1}^\top A \mathbf{1}}. \quad (7)$$

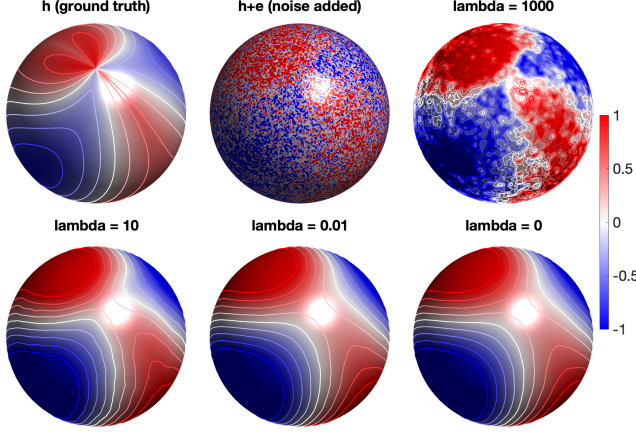


Fig. 4. The ground-truth field h_{true} (top-left) is constructed as a degree-2 spherical-harmonic mixture. A noisy observation $h_{obs} = h_{true} + \epsilon$ is generated by adding Gaussian noise $N(0, 1)$ at each vertex. The remaining panels show the recovered potentials u obtained by solving the regularized Poisson equation for different values of λ . As λ decreases, the solution approaches the classical Poisson limit and exhibits increased sensitivity to high-frequency noise, whereas larger λ suppresses fine-scale fluctuations and emphasizes the dominant low-frequency structure.

We then normalize the solution as

$$u \leftarrow \frac{\|h\|}{\|u\|} u.$$

Second, we remove any residual area-weighted mean of the potential by enforcing zero-mean condition:

$$\mathbf{1}^\top A u = 0,$$

where $\mathbf{1}$ denotes the constant vector. The Poisson equation gives solution only up to an additive constant. Thus, enforcing the zero-mean constraint provides a consistent reference level across subjects.

This global rescaling and translation preserve all spatial patterns, relative contrasts, and gradient-derived flux fields, while ensuring consistent amplitude normalization across subjects, meshes, and values of λ . All subsequent statistical analyses are unaffected by this normalization, since standard test statistics such as t - and F -statistics are invariant under global translation and scaling of the response variable [34].

3. VALIDATION

Since real cortical data do not provide a ground-truth solution for validation, we generated synthetic source fields on the unit sphere $\mathbb{S}^2$ with known analytic solutions using *spherical harmonics* $Y_{\ell m}$ [35, 36], which form an eigenbasis of the

spherical Laplacian $\mathcal{L}$ satisfying

$$\mathcal{L}Y_{\ell m} = \ell(\ell + 1)Y_{\ell m}$$

for $\ell \geq 0$ and $-\ell \leq m \leq \ell$.

We construct a noise-free ground-truth field h_0 by choosing a finite mixture of modes,

$$h_{true}(\theta, \phi) = \sum_{r=1}^R \alpha_r Y_{\ell_r, m_r}(\theta, \phi).$$

Given h_0 , the regularized Poisson equation on $\mathbb{S}^2$ admits the spectral solution (Theorem 2)

$$u_{true}(\theta, \phi) = \sum_{r=1}^R \frac{\alpha_r}{\lambda + \ell_r(\ell_r + 1)} Y_{\ell_r, m_r}(\theta, \phi).$$

This construction provides a ground-truth target u_0 for validating the cotangent finite-element solver [25, 32]. To emulate noise arising from discretization and mesh irregularity, we add a Gaussian random field $\epsilon(\theta, \phi)$ to the analytic signal and form

$$h_{obs}(\theta, \phi) = h_{true}(\theta, \phi) + \epsilon(\theta, \phi),$$

where ϵ follows the multivariate distribution $N(0, \sigma^2 I)$ over whole vertices.

In this study, we used degree-2 spherical harmonics as the ground truth (Fig. 4-top left):

$$h_{true}(\theta, \varphi) = Y_{20}(\theta, \varphi) - Y_{21}(\theta, \varphi) + Y_{22}(\theta, \varphi)$$

This gives the corresponding ground-truth

$$u_{true}(\theta, \varphi; \lambda) = \frac{1}{6 + \lambda} h_{true}(\theta, \varphi).$$

Both h_{true} and u_{true} went through the discrete zero-mean constraint

$$\mathbf{1}^\top A h_{true} = 0, \quad \mathbf{1}^\top A u_{true} = 0.$$

In the validation, no amplitude normalization with respect to h is applied to the recovered potential u . Consequently, the comparison between u and u_{true} reflects only discretization and numerical errors of the finite-element Poisson solver, without any post hoc rescaling that could artificially improve the performance.

The numerical accuracy is quantified using the area-weighted inner product (5) and the surface L^2 -norm (7). The *area-weighted correlation* between two fields f and g is computed as

$$Corr(f, g) = \frac{\langle f, g \rangle}{\sqrt{\|f\| \|g\|}}.$$

The metrics are used to quantify two complementary aspects of performance. $\|u - u_{true}\|$ (area-weighted RMSE) and

$\text{Corr}(u, u_{\text{true}})$ measures the *numerical accuracy* of the finite-element Poisson solver, i.e., how closely the recovered potential u matches the analytic ground truth u_{true} . In contrast, $\|u - h_{\text{obs}}\|$ and $\text{Corr}(u, h_{\text{obs}})$ measures the *pattern fidelity* of the recovered potential relative to the observed noisy data h_{obs} .

The validation results in Table 1 demonstrate both the numerical accuracy of the cotangent FEM discretization and the noise-filtering behavior of the regularized Poisson model. In the noise-free case ($\sigma = 0$), the recovered potential u agrees almost perfectly with the analytic ground truth u_{true} for all λ , with surface-weighted correlations exceeding 0.97 and RMSE on the order of 10^{-5} – 10^{-2} , confirming that the FEM solver accurately reproduces spherical-harmonic solutions of the Laplace–Beltrami operator. As noise is introduced, the numerical agreement between u and u_{true} remains high, indicating that discretization errors are negligible even when the input is contaminated. In contrast, the similarity between u and the noisy observation h_{obs} degrades rapidly as σ increases, especially at $\sigma = 1$, where correlations drop below 0.7, reflecting the amplification of high-frequency noise by the inverse Poisson operator. Increasing λ partially stabilizes this effect by suppressing noisy components, but also attenuates the true low-frequency signal. Overall, the table shows that discrepancies between u and h_{obs} are driven primarily by the added noise, while the FEM-based Poisson solver itself remains highly accurate across all regularization regimes.

4. RESULTS

Our primary quantity of interest is the folding-induced flow $J = -\nabla u$, which characterizes the direction and magnitude of geometric transport along the cortical surface (Fig. 3). Direct statistical inference on J is challenging because it is a vector-valued field constrained to the local tangent plane, requiring multivariate testing and estimation of cross-component covariance that varies with surface geometry. To avoid these complications, we exploit the fact that J is uniquely determined by the scalar potential u through a first-order differential operator. Consequently, testing for group differences in J is mathematically equivalent to testing for differences in the underlying potential u , since any systematic change in the flow must be reflected in the potential field that generates it. This equivalence allows us to perform statistically efficient scalar inference.

To assess group differences in cortical folding while adjusting for age and sex, we perform vertexwise linear regression on the folding potential maps $\{u_i(v)\}$, where $u_i(v)$ denotes the Poisson-derived folding potential at vertex v for subject i . At each vertex v , subject-level potential are stacked into the vector $\mathbf{u}(v) = [u_1(v), \dots, u_n(v)]^\top$ and modeled using the general linear model

$$\mathbf{u} = c_0 + c_1 \text{Group} + c_2 \text{Age} + c_3 \text{Sex} + \varepsilon, \quad (8)$$

where Group encodes diagnostic status (JME vs. control), and Age and Sex are included as nuisance covariates. The coefficient c_1 captures the local group effect on folding organization and is tested using a vertexwise t -statistic (Fig. 5-top). We also did analysis ignoring sex (Fig. 5-bottom). The results are almost undierable indicating sex is not affecting the analysis. The coefficient c_1 captures the local group effect on cortical folding organization and is assessed using a vertexwise t -statistic (Fig. 5-top). To evaluate the potential contribution of sex, we repeated the analysis excluding sex as a covariate (Fig. 5-bottom). The resulting t -statistic maps are nearly indistinguishable, indicating that sex does not influence the estimated group effect on folding organization.

Resulting p -values were corrected for multiple comparisons across the cortical surface using false discovery rate (FDR) control [37, 38]. Regions displayed in dark red or dark blue (i.e., $|t| \geq 4.3$) correspond approximately to an FDR-corrected q -value of 0.01, whereas regions shown in lighter red or blue (i.e., $|t| \geq 3.2$) correspond approximately to an FDR-corrected q -value of 0.05.

Red clusters (JME > controls) in Fig. 5 identify cortical regions in which folding potential is significantly elevated in JME. Elevated folding potential reflects stronger local curvature contrast and increased coherence of sulcal–gyral organization, indicating region-specific alterations in cortical folding geometry rather than global morphological change. These effects are spatially focal, with the strongest positive t -values concentrated along perisylvian and temporal cortices in both hemispheres, together with additional involvement of right fronto-central and medial motor regions [3, 39]. The exceptionally strong effects observed along the right Sylvian fissure may reflect a compensatory or adaptive reorganization mechanism, potentially driven by chronic epileptic activity and altered network synchronization characteristic of JME. Such hemispherically asymmetric amplification suggests that the observed effects are not spatially uniform, but instead reflect regionally specific alterations in cortical folding geometry within distributed functional networks in JME.

In contrast, blue clusters (JME < controls) in Fig. 5 indicate regions where folding potential is significantly lower in JME. These effects are more widespread and bilaterally distributed, predominantly involving the postcentral gyrus and visual cortical areas, with additional focal involvement of the right orbital gyri. Reduced folding potential reflects attenuated curvature contrast and smoother, less differentiated sulcal–gyral geometry, indicating weaker large-scale organization of cortical folding patterns. Such changes are consistent with altered maturation of sensory and association cortices and support the view that JME involves distributed network-level structural alterations rather than focal cortical abnormalities [6].

Taken together, this pattern implies that JME reflects a network-level neurodevelopmental alteration, in which some cortical regions exhibit localized structural accentua-

Table 1. Performance of the proposed Poisson flow model u on a unit sphere using degree-2 spherical-harmonic ground truth u_{true} , together with pattern similarity between the recovered potential u and the noisy observation h_{obs} under different noise levels and regularization parameters λ .

Noise	λ	$\ u - u_{true}\ $	$Corr(u, u_{true})$	$\ u - h_{obs}\ $	$Corr(u, h_{obs})$
No noise, $\sigma = 0$	1000	1.6607×10^{-5}	0.99943	1.6610×10^{-2}	0.99944
	10	5.8837×10^{-3}	0.98219	9.4085×10^{-2}	0.98219
	0.01	1.9661×10^{-2}	0.97160	1.1880×10^{-1}	0.97160
	0	1.9700×10^{-2}	0.97158	1.1883×10^{-1}	0.97158
Small noise, $\sigma = 0.1$	1000	2.2321×10^{-5}	0.99898	6.5677×10^{-2}	0.99148
	10	5.8900×10^{-3}	0.98216	1.1797×10^{-1}	0.97252
	0.01	1.9681×10^{-2}	0.97155	1.3875×10^{-1}	0.96198
	0	1.9719×10^{-2}	0.97154	1.3879×10^{-1}	0.96196
Large noise, $\sigma = 1$	1000	1.4616×10^{-4}	0.95832	6.8676×10^{-1}	0.68508
	10	6.1853×10^{-3}	0.98033	8.0396×10^{-1}	0.56841
	0.01	2.0621×10^{-2}	0.96871	8.1305×10^{-1}	0.55860
	0	2.0665×10^{-2}	0.96868	8.1307×10^{-1}	0.55858

tion while others show diffuse smoothing or reduced folding differentiation. This is consistent with models of nonuniform cortical maturation affecting large-scale brain networks, rather than focal lesions or a single localized abnormality [3, 6].

5. DISCUSSION

We compared the folding-potential findings with cortical thickness patterns to place the results in a broader morphometric context. Prior studies have reported cortical thickness abnormalities in JME. For example, reductions in cortical thickness in frontal and temporal regions have been documented relative to controls [3], while regionally increased thickness in orbitofrontal and mesial frontal cortex has also been observed [39]. Cortical thickness primarily characterizes laminar properties of the cortex, reflecting variation in the radial (normal-to-surface) direction.

Group differences in cortical thickness were evaluated by fitting the regression model (8), with age and sex included as covariates. The group effect was statistically quantified by computing the t -statistic of the group variable in the model. Statistical significance was assessed after correcting for multiple comparisons across the cortical surface using false discovery rate (FDR) control [37, 38]. Cortical thickness was computed vertexwise as the Euclidean distance between the white-matter and pial surfaces [18], yielding a local measure of laminar thickness orthogonal to the cortical sheet (Fig. 3 - bottom). Group differences in cortical thickness were assessed using the regression model (8), with age and sex included as covariates. The group effect was evaluated using the t -statistic of the group variable, and statistical significance was determined after controlling for multiple comparisons across the cortical surface using false discovery rate (FDR) correction [37, 38]. The resulting t -statistic maps are shown

in Fig. 6. Analyses adjusted for age and sex and those adjusted for age alone yielded nearly identical spatial patterns, indicating a negligible contribution of sex to the observed group differences.

Red clusters ($JME > controls$) indicate regions of increased cortical thickness in JME, predominantly involving bilateral temporal and perisylvian cortices, with additional distributed effects across frontal gyri. In contrast, blue clusters ($JME < controls$) indicate regions of reduced cortical thickness, primarily confined to primary sensorimotor and visual cortices. Overall, JME is characterized by a predominance of regionally increased cortical thickness, with more localized thinning in sensory and visual areas. Darker red or blue ($|t| \geq 4.1$) correspond to an FDR-corrected q -value of approximately 0.01, whereas lighter color intensities ($|t| \geq 2.8$) correspond to approximately 0.05.

As expected, the cortical thickness findings only partially overlap with the folding-potential results, underscoring that these measures capture distinct yet complementary aspects of cortical organization. Cortical thickness primarily reflects laminar architecture along the radial direction, whereas the folding potential encodes tangential geometric organization and curvature contrast along the cortical surface. Regions exhibiting altered folding potential do not uniformly coincide with regions showing cortical thickness differences, indicating that changes in folding organization cannot be explained solely by laminar thickening or thinning. This dissociation was further examined using partial correlation between folding potential and cortical thickness, controlling for age and sex (Fig. 7). We observed little to no significant correlation in regions exhibiting group effects, confirming that folding potential and cortical thickness capture largely independent aspects of cortical organization. This dissociation suggests that folding potential captures additional reorganization of sulcal-gyral patterning that is not accessible through con-

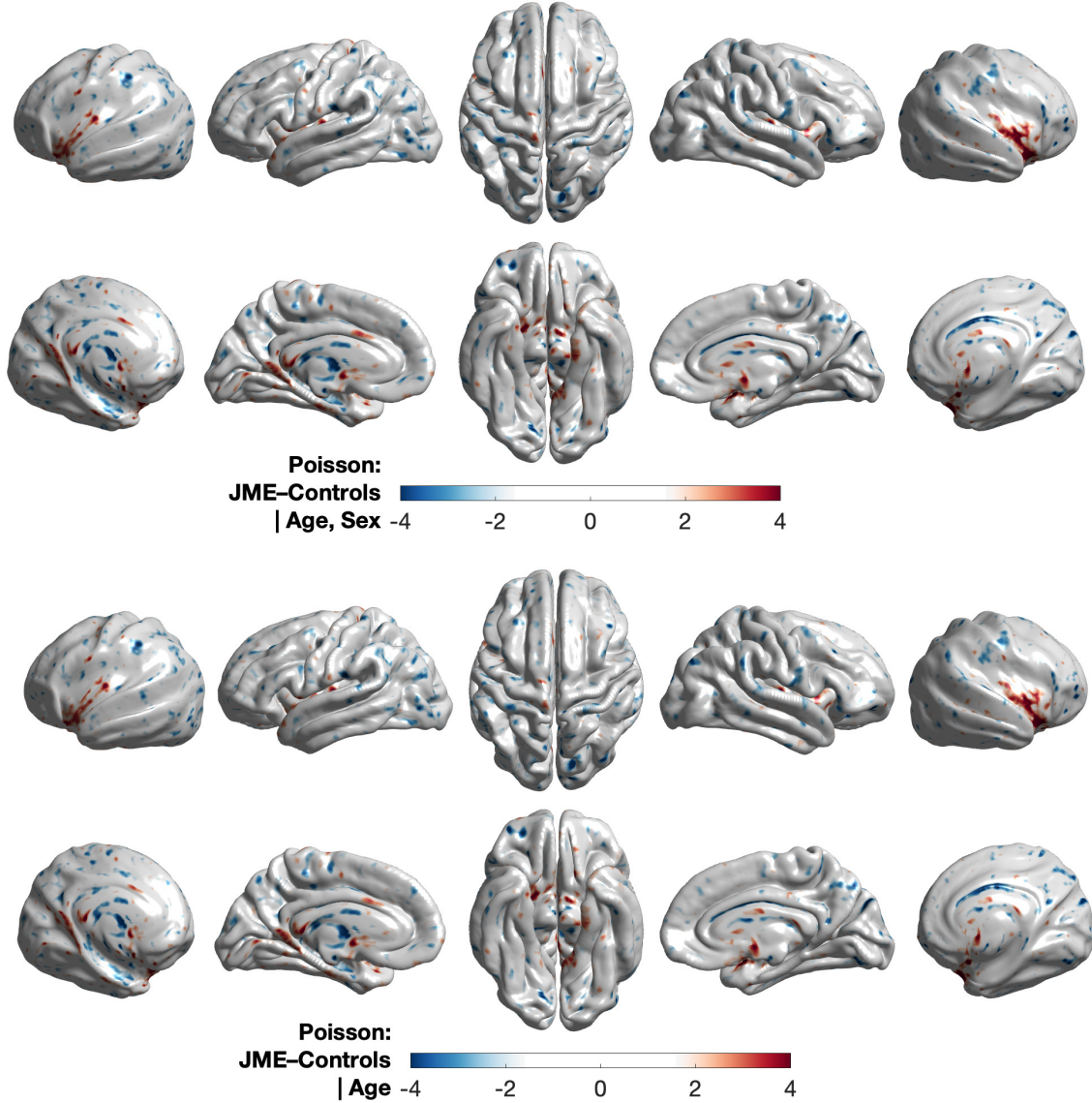


Fig. 5. t -statistic maps for the JME-control contrast, adjusted for age and sex (**top**) and for age only (**bottom**). The far left and far right columns are flattened surface. There is a negligible contribution of sex to the group effect. Red clusters (JME>controls) exhibit dominant increases in folding potential over left fronto-central and medial motor regions, with more focal right-hemisphere effects along the Sylvian fissure, indicating asymmetric, network-level alterations in cortical organization in JME.

ventional thickness analysis alone [40].

6. CONCLUSION

This study introduces a geometry-driven Poisson flow framework for modeling and analyzing cortical folding organization, and applies it to JME. By recasting cortical folding as a source-sink system derived from mean curvature and solving a regularized Poisson equation on the cortical manifold, we obtain a smooth scalar potential whose gradient defines an in-

trinsic folding-induced flow. This provides a principled link between local folding polarity and large-scale geometric organization, while avoiding the instability and interpretational ambiguity associated with direct vector-field inference.

Applied to JME, our results reveal a strikingly asymmetric and bidirectional pattern of cortical folding alterations. Folding geometry is known to be influenced by axonal tension, differential growth, and network connectivity [1, 2]. Enhanced sulcal depth in motor regions may indicate atypical constraints or delayed pruning within corticothalamic-motor

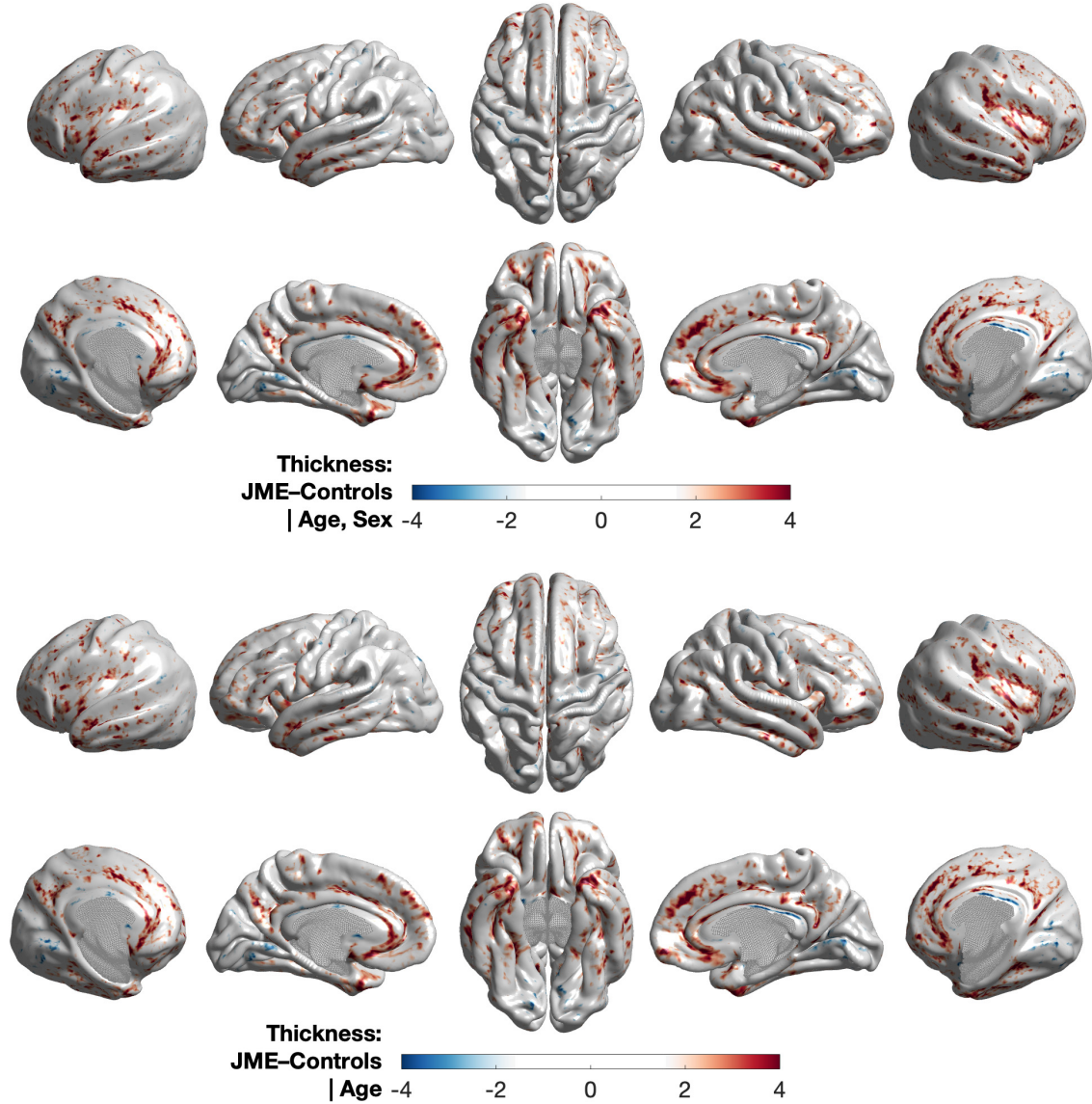


Fig. 6. t -statistic maps of cortical thickness for the JME-control contrast, adjusted for age and sex (**top**) and for age only (**bottom**). The two analyses yield nearly identical results, indicating a negligible contribution of sex to the group effect. Red clusters (JME>controls) denote regions of increased cortical thickness in JME, most prominently involving bilateral temporal and perisylvian cortices, with additional distributed effects across frontal gyri. Blue clusters (JME<controls) indicate regions of reduced cortical thickness, primarily localized to central gyri and visual cortical areas. Overall, JME is characterized by a predominance of regionally increased cortical thickness relative to controls.

circuits, while reduced sulcal differentiation in association cortex may reflect more heterogeneous or less synchronized maturation. Such structural reorganization provides a plausible substrate for the widespread functional hyperexcitability and network-level abnormalities characteristic of JME.

Future work should integrate the present framework with functional imaging to directly relate folding geometry to seizure susceptibility and cognitive phenotype. The regularization parameter λ governs the spatial scale of the inferred

patterns; while fixed here for consistency, adaptive or multi-scale extensions may further improve sensitivity. In addition, longitudinal data will be critical to determine whether the observed folding differences reflect altered developmental trajectories rather than static group effects. These directions are left for future study.

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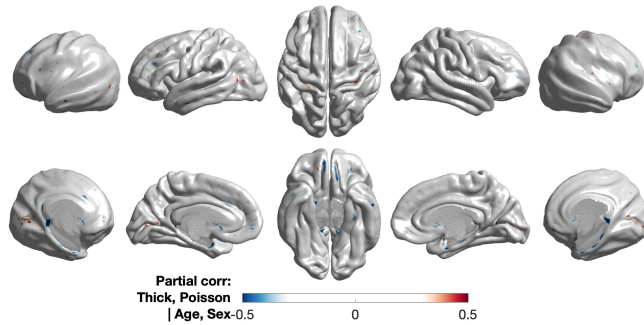


Fig. 7. Partial correlation between folding potential and cortical thickness, adjusted for age and sex. Most cortical regions show little to no partial correlation, indicating minimal statistical association between the two measures. This suggests that folding potential captures aspects of cortical organization that are largely independent of cortical thickness.

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